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Eeg based smart emotion recognition using meta heuristic optimization and hybrid deep learning techniques

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In the domain of passive brain-computer interface applications, the identification of emotions is both essential and formidable. Significant research has recently been undertaken on emotion identification with electroencephalogram (EEG) data. The aim of this project is to develop a system that can analyse an individual's EEG and differentiate among positive, neutral, and negative emotional states. The suggested methodology use Independent Component Analysis (ICA) to remove artefacts from Electromyogram (EMG) and Electrooculogram (EOG) in EEG channel recordings. Filtering techniques are employed to improve the quality of EEG data by segmenting it into alpha, beta, gamma, and theta frequency bands. Feature extraction is performed with a hybrid meta-heuristic optimisation technique, such as ABC-GWO. The Hybrid Artificial Bee Colony and Grey Wolf Optimiser are employed to extract optimised features from the selected dataset. Finally, comprehensive evaluations are conducted utilising DEAP and SEED, two publically accessible datasets. The CNN model attains an accuracy of approximately 97% on the SEED dataset and 98% on the DEAP dataset. The hybrid CNN-ABC-GWO model achieves an accuracy of approximately 99% on both datasets, with ABC-GWO employed for hyperparameter tuning and classification. The proposed model demonstrates an accuracy of around 99% on the SEED dataset and 100% on the DEAP dataset. The experimental findings are contrasted utilising a singular technique, a widely employed hybrid learning method, or the cutting-edge method; the proposed method enhances recognition performance.

Keywords Electroencephalogram, Electroculogram, Artificial Bee colony, Grey Wolf Optimizer, Hybrid learning methods, Convolutional neural network

Over the last decade, there has been some focus attached to identifying human emotions, especially due to advances made in artificial intelligence. It is therefore important to accentuate the need to recognize human emotional states toward enhancing natural and intelligent interactions between man and machine. Globally, emotions are an important constituent of human existence and they influence every aspect of human life including interactions, participation, and training[1–4]. Although there has been extensive research on this subject, the modern approaches such as facial movements, gestures, and voice control often fail to meet the precise requirement for the identification of feel.

It is possible to consider EEG-signals, providing accurate information about the subject's status disregarding appearance and behavior. However, modern EEG-based emotion detection algorithms face significant challenges including EMG/EOG contamination, limited power of feature extraction and poor accuracy. Other indexes such as fMRI[5,6], PET, EEG, or MEG are tangible and can provide direct and clear responses. Compared to other physiological assessments, access to EEG has been easier which has made the assessment the preferred tool among scientists to capture recognition of emotions[7]. Furthermore, EEG signals posses certain characteristics

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including: simplicity and increased susceptibility to external stimuli[7]. Stored patterns within cortex of the brain is most easily recorded when a person is least active and has their eyes closed[2]. The trends are estimated by comparing intensity range of 0.5 to 100 V, that is around 1000 times lower when compared to the intensity range of EEG waves[2]. Various frequencies sets are being employed to classify human cerebral waves:

delta (0.1–4) Hz[8], theta (4–8) Hz[8], alpha (8–13) Hz[8], beta (13–30) Hz[8], and gamma (30–64) Hz[8].

The field of emotional modeling is primarily based on two basic traits, namely valence and excitation. The concept of valence refers to range of good or negative emotions that a human being experiences in response to a stimulus. Arousal, on the other hand, signifies the continuum of alertness or responsiveness to stimuli, which can vary from a passive to an active state, both physiologically and psychologically. In the context of electroencephalography (EEG), these emotional dimensions are mapped through various brain regions, as depicted in Fig. 1. These regions encompass the fronto-polar (Fp), temporal (T), central ©, frontal (F), parietal (P), and occipital areas[9]. Specifically, electrodes Fp1 and Fp2 are positioned on left and right frontal lobes, respectively. The montage channels AF3 and AF4 are located centrally between the eyes. Additional electrodes, namely FC5[9], FC1[9], FC2[9], FC6[9], C3[9], C4[9], CP5[9], CP1[9], CP2[9], and CP6[9], are strategically placed across the scalp. The O1 and O2 regions are situated above the primary visual cortex, as demonstrated in Fig. 1(a). The valence-arousal dimensional modelling of emotion, widely utilized in numerous research studies, is exemplified in Fig. 1 (b).

This EEG-based approach offers an alternative to the traditional, and often more challenging, behavioral assessments conducted by clinicians. To comprehend a patient's emotional state, medical professionals typically require extensive knowledge and expertise. However, with evolution of brain cortex-computing interface technologies and neuroimaging, acquisition of brainwave data has been simplified. Current advancements enable the practical[10] and emotional[11] control of devices, such as mobility aids[11], communication tools[12], and prosthetic limbs[13,14], through wearable EEG headsets[10], a feat previously unattainable. This study employs various machine learning techniques to analyze EEG data, aiming to detect and categorize a wide array of human emotions, thereby facilitating the interpretation of emotional states from EEG waves.

Research problem and objectives

In the domain of passive brain-computer interface applications, the identification of emotions is both essential and difficult. Extensive research has been undertaken on emotion identification with electroencephalogram (EEG) data. Nonetheless, there is an ongoing demand for enhanced precision and efficacy in this domain. This research offers multiple innovative contributions to the domain of emotion identification through EEG data.

- Artifact elimination by Independent Component Analysis (ICA) facilitates the removal of artefacts from Electromyogram (EMG) and Electrooculogram (EOG) signals in EEG recordings, hence improving the quality of EEG data.
- Hybrid Meta-Heuristic Optimisation Algorithm (ABC-GWO): The application of a hybrid meta-heuristic optimisation approach for feature extraction, integrating the advantages of the Artificial Bee Colony (ABC) and Grey Wolf Optimiser (GWO) algorithms. This innovative method markedly enhances the precision of emotion recognition.

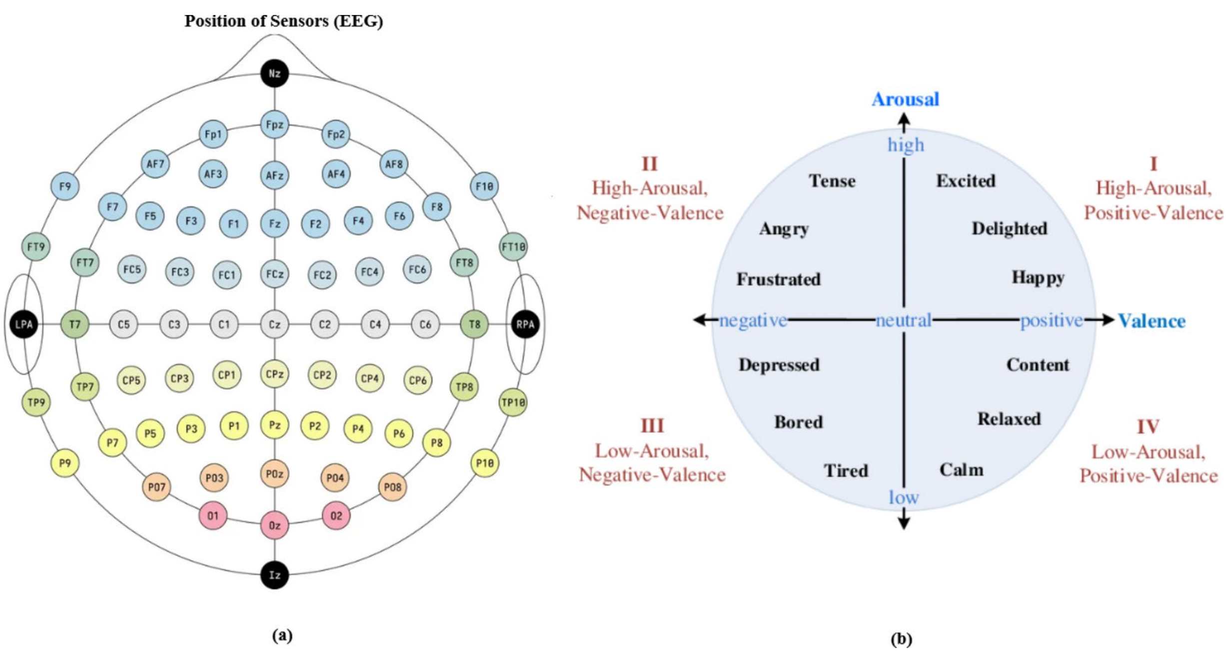


Fig. 1. (**a**) Electrodes Position (**b**) Two-dimensional valence arousal space of emotion.

- The suggested methodology undergoes comprehensive evaluation on two publicly accessible datasets (DEAP and SEED), exhibiting enhanced performance relative to previous methods, with accuracy attaining 100%.

Research gap

Promising results were obtained in recent research on EEG-based emotion recognition but quite commonly suffers greatly from limitations regarding accuracy and robustness. Most approaches fail to clean enough noise and artifacts from EEG signals, which directly affects the classification of emotions with incorrect data. Furthermore, conventional feature extraction methods often fail to adequately integrate the complex patterns in EEG data required for accurate emotion recognition. This study bridges the limitations by including artifact removal in detail and a highly advanced feature extraction method, which would significantly enhance the effectiveness and reliability of an EEG-based emotion identification system. Here, these issues are systematically carried out to improve this area of science of emotion recognition with a more accurate and reliable approach that generalizes to real-life scenarios, such as human-computer interaction and mental health diagnosis with adaptive learning environments.

The subsequent parts of document are prepared as trails: Sect. 2 focuses on an extensive synthesis of articles relevant to the topic at hand. Section 3 presents the DEAP[15] and SEED[16] databases and defines experiment configurations used. Moreover, the paper provides a novel hybrid-learning model specifically for emotion detection and describes its major components like pre-processing, feature reduction, and validation techniques. It also gives a brief idea about the working of Convolutional NeuralNetworks (CNN), ArtificialBeeColony (ABC), and GreyWolfOptimizer (GWO). In the following Sect. 4, a complete appraisal of experimental outcomes that resulted from deploying different models for DEAP[15] and SEED[16] datasets. Lastly, the last sub-section of this section is the discussion and conclusion of the study[17].

Related works

Emotion recognition has recently received increased attention because of its associations with various disciplines including psychology, physiology, learning studies, marketing, and healthcare. It has become common to classify emotions which has given rise to different classification methodologies. These can be broadly categorized into three main approaches: Supervised MLearn Models, Unsupervised MLearn Models, Reinforcement MLearn Models, DLearn Models, and EnLearn Models[18–22].

Conventional machine learning approaches

Previous approaches for emotion recognition mainly rely on feature extraction and traditional classifiers. For example, Wang et al. (2020)[23] compared the results of implementing SVM and CNN algorithms (LeNet and ResNet) in SEED and MAHNOB-HCI databases. It was established by such studies that greater models could produce better outcomes particularly when adopting data enrichment approaches. However, studies may suffer from low statistical power due to small sample sizes or low variety of participants which complicates generalization of the results[18,19].

Deep learning techniques

In the past few years, deep learning (DLearn) has taken emotion recognition to the next level by automating feature extraction and selection process. Different architectural styles include the Convolutional NeuralNetwork (CNN)[20], DeepBelief Network(DBN)[24], Graph CNNs, Long ShortTerm Memory (LSTM) networks[23] and Capsule Networks (CN)[26]. Pandey et al., (2021) proposed a CNN-based model using scalogram images of EEG, which are tested on DEAP dataset with a median precision rate of 54% and has a lower success rate while cross-checking with the SEED dataset[27]. Yucel Cimtay et al. (2020) also proposed a pretrained CNN architecture and obtained an average recognition rate of 58.10% of DEAP database through training with SEED database[28]. Such research suggests improvement but points to problems of overfitting and relatively poor transfer between different sets.

Hybrid approaches and multimodal techniques

Hybrid models, incorporating a combination of several algorithms, have been recently shown to have potential in higher performance in emotion recognition. One promising algorithm, called BimodalLSTM, takes input from both EEG as well as peripheral physiological inputs and was found to be capable of solving the issues regarding emotion recognition using a multimodal paradigm by Tang et al. (2021).

Many researchers have addressed specific emotional states. Asma Baghdadi et al. (2020) had proposed “DASPS: A Database for Anxious States based on a Psychological Stimulation,” using a stacking sparse autoencoder algorithm, with Hjorth variables as its features, reaching mean precision rates of 83.50% and 74.60% for different levels of stress[30,31]. Bachmann et al. introduced a feature extraction approach that applied Adaptive Multiscale Entropy for the detection of driving fatigue with SVM classifiers, obtained a detection rate of 95%[31]. Alakus et al. have also done research on their own dataset named GAMEEMO by applying spectral entropy with the help of a bidirectional LSTM, with an average precision of 76.91%[32]. An et al. have used deep CNN (DCNN) and Conv-LSTM models for emotion recognition later in 2021[33].

Rajpoot et al. (2021) carried out a subject-invariant experiment based on the use of LSTM with Channel Attention autoencoder and CNN with Attention model on the DEAP, SEED, and CHBMIT datasets[34]. Jingcong Li et al. (2021) used graph neural network with mean precision rates of 86.8% and 75.27% on the SEED and SEED-IV datasets, respectively[35]. Features such as mean absolute deviation and standard deviation are used with SVM, LDA, ANN, and k-NN classifiers, resulting in a high level of classification accuracy in a subject-dependent methodology by Rahman et al. (2020)[36]. Dong et al. (2020) utilized DNNs for the recognition of emotions through EEG waves recorded from the DEAP dataset and showed that DNNs could handle a huge amount of varied training records[37].

Recent innovations

Recently new methods have been proposed, which involve synchrosqueezing wavelet transform maps in conjunction with deep architectures such as ResNet-18[38]. Continuous wavelet transform and transfer learning based on CNNs have also attained higher accuracy in emotion recognition[39]. Hybrid models, which are based on wavelet convolutional neuralnetworks combined with SVM, have established robust frameworks for emotion recognition[40]. An ensemble deep learning approach including effective connectivity maps integrated from the brain produced 96.67% accuracy in emotion recognition, as proposed by Bagherzadeh et al. (2021).

Critical analysis and relevance

The traditional and deep learning methods have impacted the domain highly, but they are sensitive to flaws in datasets, overfitting, and improper handling of artifacts within the EEG signals. This limitation of application to broader, real-world scenarios comes from the dependence of the methods on limited and homogeneous datasets. In the large number of studies, robust artifact removal techniques are generally not provided, and noisy data may be left behind, compromising the accuracy of the emotion recognition systems.

Significance of the proposed methodology

The present study overcomes the shortcomings of previous works by using ICA for artifact removal and a hybrid meta-heuristic optimization algorithm (ABC-GWO) for feature extraction. Thus, the quality of the EEG data improves and the emotion recognition systems' accuracy increases up to 100% on the DEAP dataset and 99% on the SEED dataset[15,16]. It is in this line that the paper fills gaps left behind by previous work and whereby the proposed methodology is presented here as a new approach.

Methods and materials

Dataset description

The present study utilizes the DEAP[15] and SEED[16] datasets, which are widely recognized as benchmarks in the field of emotion identification research.

1. *SEED Dataset*: The SEED dataset had a crucial role in evoking emotional responses from participants using short films as a medium as mentioned in Table 1. These films, originating from China, were each accompanied by a musical score to enhance the emotional experience. The design of the experiment aimed to simulate a naturalistic setting, thereby provoking significant physiological reactions. A total of fifteen clips, averaging four minutes in length, were utilized per session. The dataset encompasses estimations of happiness, sadness, and neutrality, with each emotional category represented by five distinct film clips. Participants underwent a structured session that included a five-second preview of each clip, followed by a 45-second reflection period and a 15-second intermission for recuperation. The emotional states of the subjects were assessed using self-report questionnaires. The group comprised Chinese pupils who claimed to have normal eyesight and hearing to themselves. Each participant engaged in experimental procedure three times, with a minimum one-week interval between sessions. The EEG data has been captured employing a sixtytwo-channel electrode in accordance with the 10–20 system, sampled at 1000 Hz, filtered between 0.5 and 70 Hz, and subsequently downsampled to 200 Hz for signal processing.
2. *DEAP Dataset*: The DEAP dataset stands as a pivotal resource for emotion recognition via EEG signals. It comprises recordings from 32 participants who were monitored while viewing music videos (Table 1). Each participant's neural activity was recorded over 40 video sessions, using a 40-channel setup with continuous referencing, initially sampled at five hundred and twelve Hz and later processed to hundred and twenty eight Hz. The study involved exposing the subjects to a sequence of video clips, wherein each clip was marked with four distinct emotional descriptors: arousal, valence, dominance, and like/dislike. The participants' assessments on valence, arousal, dominance, and like/dislike for each stimulus were captured using self-assessment questionnaires. The utilization of the DEAP dataset enabled the delineation of nine distinct emotional states across several dimensions, including joy, tranquility, anxiety, excitement, apathy, and melancholy.

Preprocessing

Filtering

In the field of electroencephalography (EEG), different types of background noise make it necessary to preprocess raw data in order to improve the quality of EEG readings. Delta (0–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz) are four main rhythms that make up EEG. Other frequencies can go up to 50 Hz. These beats,

Dataset	Name of the array	Size of the array	Contents in the array
SEED Dataset	EEG Data	$18 \times 62 \times 48{,}000$	(videos/trials) x channels x samples
	Labels	15×3	(videos/trials) x labels
DEAP Dataset	EEG Data	$40 \times 40 \times 8064$	(videos/trials) x channels x samples
	Labels	40×4	(videos/trials) x labels

Table 1. Database structure.

which can be seen in a medical setting, tell us a lot about the most basic EEG patterns. To get features out of data in the suggested study, a special DualTree complexwavelet transform is used through a Hanningwindow. Based on the ValenceArousal Two-Dimensional Approach, these traits are then marked as either good, negative, or neutral. A rating above the average indicates a high arousal/valence class (positive, 1), while a rating below the median suggests a low arousal/valence class (negative, 1). Conversely, a rating equal to or less than median signifies neutral arousal/valence label (0, 0). To assess emotion recognition capabilities, a binary categorization task was designed.

Preprocessing of raw data is essential to improve the quality of EEG signals. Methods such as IndependentComponent Analysis (ICA), specifically rapid ICA restriction method, are commonly employed to semiauto-matically eliminate EOG and EEG grids from EEG recordings, facilitated by software applications.

As an initial step in signal processing, filtering is utilized, followed by a decomposition into sub-bands using the discrete cosine WaveletTransform (DT-CWT). Filtering can be mathematically illustrated by a convolution operation. For a provided signal z(t) and filter f(t), the filtered signal y(t) can be expressed as in (1).

$$y\left(t\right)=\int_{-\infty}^{\infty}z\left(\tau\right).f\left(t-\tau\right)d\tau \tag{1}$$

DualTree complexwavelet transform (DT-CWT)
The DualTree ComplexWavelet Transform (DT-CWT) uses a dualtree of wavelet filters to tell the difference between the real and fake parts of complex coefficient-ts. Compared to the DiscreteWaveletTransform (DWT), the DTCWT does a better job of preventing aliasing and keeping the estimated shift invariance. The DTCWT is represented as a series of convolution operations with wavelet based filters. For an input signal z(t), and $\Psi_{j,k}$ as the wavelet basis functions, the DTCWT coefficients are calculated as in Eq. (2).

$$W_{j,k}=< f,\,\Psi_{j,k}> \tag{2}$$

Independent component analysis (ICA)
If the accuracy of EEG data need to be improved before they are used, IndependentComponentAnalysis (ICA) is utilized. In particular, rapid ICA restriction method is used to get rid of flaws from electrooculogram (EOG) and electromyogram (EMG) records that are in EEG band. ICA is employed to remove artifacts from EMG and EOG in EEG recordings, improving data quality. The fastICA algorithm parameters that are used are as follows:

- Maximum number of iterations: 200.
- Tolerance: 1e-4.

Hybrid ABC-GWO based feature extraction
In this project, obtaining features from EEG data channels is done using a hybrid method that includes ABCGWO optimization technique. This new method takes best parts of both ABC and GWO techniques and uses them together to quickly pick out or derive beneficial characteristics from EEG information. The hybrid ABCGWO algorithm uses combined ingenuity of all these optimization approaches to find most useful features that describe mental states or actions that are captured in EEG waves. A hybrid meta-heuristics optimization algorithm (ABC-GWO) is used for feature extraction. This novel approach combines the strengths of the Artificial Bee Colony (ABC) and Grey Wolf Optimizer (GWO) algorithms, significantly improving the accuracy of emotion detection. The detailed workflow of the Hybrid ABC-GWO algorithm is represented in Fig. 2.

ABC optimization technique

The ABC optimization technique is used to pull out pertinent characteristics that successfully differentiate between different brain processes or stages in EEG analysis. A number of important steps can be taken to break down the procedure:

- Initialization Stage: The process starts by making a bunch of random responses, each represented by letter X_{ij} and containing a possible set of attributes drawn from EEG waves. It is important to note that these answers are based on problem's size and take into consideration lower (LB_j) and higher (UB_j) limits for every parameter. Also, the AbandonmentCounter (AC_i) for every option is lowered to zero to see how well it works.

$$X_{ij}=\,LB_j\,+\,\lambda\,\left(UB_j-LB_j\right) \tag{3}$$

Where:

- X_{ij} denotes the jth parameter of the ith employed bee's solution
- LB_j denotes lower bound of jth parameter
- UB_j denotes upper bound of jth parameter
- λ is a random number within the [0,1] range
- Phase of Employed Bees: During this stage, modifications are made to the existing solutions of employed bees to generate candidate solutions. Each candidate solution, denoted by V_{ij}, is formulated by updating a single parameter of the solution $X_{ij}$of the employed bee.

$$V_{ij}=\,X_{ij}+\,\varnothing\,\left(X_{kj}-\,X_{ij}\right) \tag{4}$$

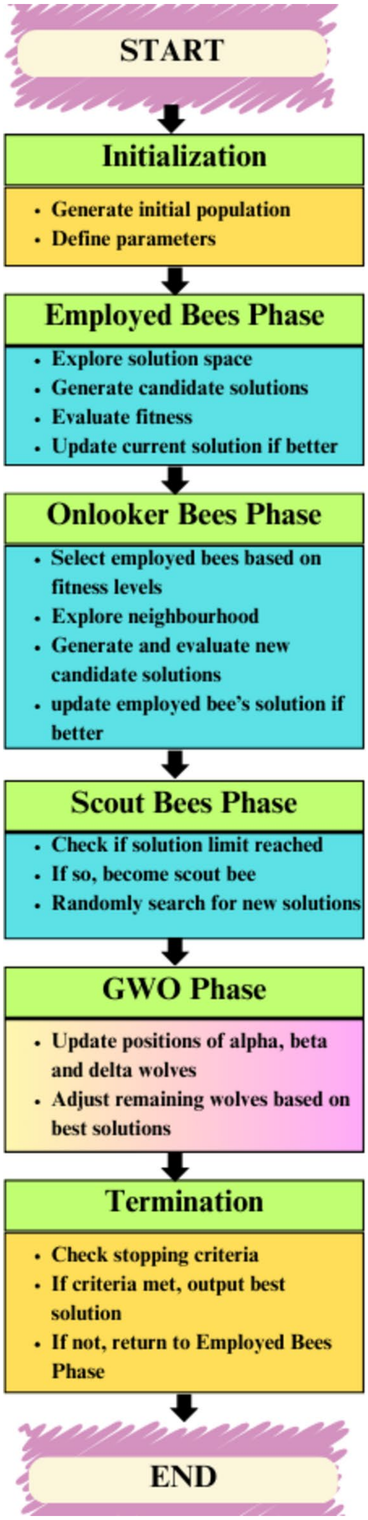


Fig. 2. Hybrid ABC-GWO algorithm workflow.

Where.

- V_{ij} denotes jth dimension of ith candidate solution
- X_{ij} denotes jth dimension of ith employed bee's solution
- X_{kj} denotes jth dimension of kth employed bee's solution
- $\varnothing$ is a random number within the $[-1, +1]$ range

- Evaluation of Fitness: The fitness for each potential solution is assessed based on the specific objective function of the problem. This function provides a metric for evaluating the effectiveness of the candidate solution in extracting pertinent data from the EEG signals.

$$Fitness(V_{ij}) = \frac{1}{1 + f(V_{ij})} \tag{5}$$

Where.

- $Fitness(V_{ij})$ denotes fitness value of ith candidate solution
- $f(V_{ij})$ denotes objective function value of ith candidate solution
- Phase of Employed Bees: At this stage, possible solutions are thought up by changing hired bees' current approaches. By changing just one variable of answer given by employed bee, every possible solution is created. An equation that uses a random choice of a nearby solution from group of hired bees makes this adjustment easier. The problem's particular objective work is used to judge usefulness of each possible answer.
- Phase of Onlooker Bees: In order to optimize their responses, onlooker bees select hired bees according to fitness rankings. The process of screening is conducted by utilization of a roulette wheel process, wherein the probability of selecting a bee that is used is directly correlated with its level of fitness. The response proposed by selected employed bee is subsequently refined by spectator bee. When new solution outperforms employed bee's answer, it is replaced; if not AbandonmentCounter of employing bee is increased.
- Phase of Scout Bees: When the desertion counters of employed bees exceed a specified threshold, they undergo a transformation into scout bees. The process of problem exploration involves generation of arbitrary solutions across dimensions of challenge by scout bees. This investigation facilitates prevention of population stasis among hired bees, hence promoting variation in pursuit of ideal characteristics.

Within the realm of EEG signal analysis, every approach represents a possible collection of characteristics derived from EEG domains. The aforementioned qualities may include spectrum power, consistency, entropy measurements, along with other relevant indicators of neural activity. Every solution's efficiency is evaluated by its ability to accurately capture distinguishing characteristics in EEG data, including distinguishing among various states of thought or identifying aberrant brain activities. The ABC technique's repeated optimization procedure, guided by assessment of fitness principles, enables detection of informative characteristics from EEG data. The ABC method, as suggested, is employed for obtaining elements from EEG signals, aiming to uncover complicated patterns and intrinsic characteristics of brain function. The aforementioned optimization technique enables collection of traits including spread of PowerSpectral density (PSD) throughout different frequencies (delta, theta, alpha, beta, gamma), either absolute or comparative strength inside particular frequency bands, chronological traits like average, variance, skew, kurtosis, and waves shape, spectral parameters that involve frequency entropy, spectrum edge number, and spectrum centroid, time-frequency illustration using wavelets and shorttime Fourier shifts potentials associated with events (ERPs) like P300 or N400 factors, operational connectivity statistics like coherence, stage synchronization, and correlations, intricacy measures like aentropy, fractals dimension, and LempelZiv intricacy, and spatial properties includes. This comprehensive examination enables a more profound comprehension of brain functionality, cognitive mechanisms, and neurological conditions, consequently augmenting comprehensibility and diagnostic efficacy of EEG data.

GWO optimization technique

The GWO algorithm by analogy with the social hierarchy and hunting techniques of grey wolves is recommended for the feature extraction of EEG signals. The GWO mimics wolves hunting aspects as the key foragers negotiate the intricate terrain of the EEG data. Initially, a set of potential solutions, symbolized as wolf packs, is randomly created. Within this set, four unique groups are recognized: alpha (α), beta (β), delta (δ), and omega (ω). As the iterations progress, the α, β, and δ wolves embody the optimal solutions, steering the optimization process, while the wolves surround them to probe for alternate solutions. Mathematical equations dictate the encirclement process are represented below that, modifying the positions of ω wolves in relation to α, β and δ.

$$\vec{E} = |\vec{C}.\vec{Y_p}(t) - \vec{Y}(t)| \tag{6}$$

$$\vec{Y}(t-1) = \vec{Y_p}(t) - \vec{A}.\vec{E} \tag{7}$$

In this equation, t will denote the current iteration, $\vec{C} = 2.\vec{k_2}$, $\vec{A} = 2.\vec{b}.\vec{k_1}.\vec{b}$, $\vec{Y_p}$, will denote the prey's position vector, $\vec{Y}$ will denote a grey wolf's position vector, $\vec{b}$ will experience a gradual decrease from 2 to 0, while k_1 and k_2 will denote random numbers over the [0, 1] range.

In the mathematical representation of the hunting behavior of grey wolves, the GWO algorithm consistently assumes α, β and δ to possess superior knowledge of the prey's location (optimum). Consequently, the positions of the top three solutions (α, β and δ) acquired thus far are preserved. Furthermore, the remaining wolves (ω) are required to adjust their positions in relation to α, β and δ. The subsequent equations (8 to 10) delineate the mathematical framework for the repositioning of the wolves:

$$\vec{E_\lambda} = |\vec{C_1}.\vec{Y_\alpha} - \vec{Y}|$$

,

$$\overrightarrow{E_\beta} = \left|\overrightarrow{C_2}.\overrightarrow{Y_\beta} - \overrightarrow{Y}\right|$$

,

$$\overrightarrow{E_\delta} = |\overrightarrow{C_3}.\overrightarrow{Y_\delta} - \overrightarrow{Y}| \tag{8}$$

$$\overrightarrow{Y_1} = \overrightarrow{Y_\alpha} - \overrightarrow{B_1}.\left(\overrightarrow{E_\alpha}\right),$$

$$\overrightarrow{Y_2} = \overrightarrow{Y_\beta} - \overrightarrow{B_2}.\left(\overrightarrow{E_\beta}\right),$$

$$\overrightarrow{Y_3} = \overrightarrow{Y_\delta} - \overrightarrow{B_3}.\left(\overrightarrow{E_\delta}\right) \tag{9}$$

$$\overrightarrow{Y}(t+1) = \frac{\overrightarrow{Y_1} + \overrightarrow{Y_2} + \overrightarrow{Y_3}}{3} \tag{10}$$

In these mathematical formulations, the current solution's position is represented by $\overrightarrow{Y}$, the iteration count is symbolized by *t*, $\overrightarrow{Y_\alpha}$ positions of alpha, beta, and delta are denoted by $\overrightarrow{Y_\beta}$, $\overrightarrow{Y_\delta}$ and respectively, while $\overrightarrow{C_1}$, $\overrightarrow{C_2}$, $\overrightarrow{C_3}$ and $\overrightarrow{B_1}$, $\overrightarrow{B_2}$ and $\overrightarrow{B_3}$ are all vectors of random values.

These modifications maintain a balance between the exploration and exploitation of the solution space, aided by vectors $\overrightarrow{B}$ and $\overrightarrow{C}$, which manage the exploration-exploitation equilibrium. Specifically, $\overrightarrow{B}$ progressively diminishes over iterations, assigning exploration and exploitation stages, while $\overrightarrow{C}$ determines the extent of exploration or exploitation. By amalgamating these mechanisms, the GWO algorithm effectively traverses the EEG data landscape, extracting features such as power spectral density (PSD) across frequency bands, time-related attributes, spectral characteristics, event-related potentials (ERPs), functional connectivity metrics, complexity measures, and spatial features. This methodology amplifies the comprehension of brain activity patterns, cognitive processes, and neurological disorders, augmenting the interpretability and diagnostic precision of EEG data analysis.

Proposed hybrid GWO-ABC algorithm

The Hybrid ArtificialBee Colony-GreyWolf Optimizer system is a unique algorithm made specifically for feature extraction of Electroencephalogram signals. One possible solution to this problem is the use of the ABC-GWO algorithm; the approach is considered optimal because it combines adaptive search processes derived from the ABC and GWO algorithms. While working with EEG feature extraction spheres, this type of ABCGWO behaves in a varied way, which allows it to operate at lower levels in the extensive search space. ABCGWO imitates the foraging behavior of bees and the social ranking system of grey wolves to guide the optimization of natural intelligence-based systems. It uses a systematic technique to determine the desired feature set. This is how it works:

1. *Phase of Initialization*: By generating a population of possible solutions, each of which corresponds to a feature set extracted from EEG signals, the algorithm is thus kicked off. Each feasible solution is a feature vector, which is a transformation of the EEG data with its dimensions being the numerous aspects of brain activity.
2. *ABC Phase*: The algorithm employs the employed and onlooker bees to explore and exploit possible feature sets. The bees convert these values gotten from the EEG data and continually adjust them in an attempt to further improve their performance. The employed bees exploit promising feature values and solutions with these values, while onlookers also exploit the solutions based on their performance.
3. *GWO Phase*: Firstly, this grey wolf optimization phase enhances the ABC model by optimizing grey wolf hunting actions to fine-tune the feature sets. Grey wolves modify the value of characteristic vectors based on the optimal output solution. Alpha, beta, and delta wolves steer the optimization process, while omega wolves surround and refine the solutions to enhance their quality.
4. *Phase of Iterative Refinement*: The method continuously improves feature sets obtained from the EEG input during iterative optimization phase. The system adjusts settings and chooses characteristics according to their efficacy, gradually approaching an optimal collection of characteristics.
5. *Evaluation and Selection Phase*: During each iteration, the algorithm evaluates the effectiveness of the feature sets based on predetermined metrics. Features that are most proficient in distinguishing between different brain states or activities are kept, while features with lower informational value are eliminated.
6. *Convergence and Output Phase*: The algorithm continues until it satisfies the convergence conditions, which could be a set limit of iterations or achieving a desired level of performance.

The resulting feature set from the ABC-GWO algorithm efficiently captures significant patterns and dynamics within the EEG signals. The parameter settings utilized for proposed ABC-GWO are listed below:

- Artificial Bee Colony (ABC) Parameters:
  - Number of employed bees: 50.
  - Number of onlooker bees: 50.
  - Limit for scout bees: 100.

  - Maximum number of iterations: 200.

- Grey Wolf Optimizer (GWO) Parameters:

  - Number of wolves (search agents): 30.
  - Maximum number of iterations: 200.
  - Coefficients *a*, *A*,*C*: dynamically updated, with a decreasing linearly from 2 to 0, while *A* and *C* are random vectors in the range [0, 2].

Through the integration of ABC and GWO algorithms, the hybrid method effectively detects optimal characteristics within EEG data, ultimately enabling improved examination and understanding of brain function. By implementing this methodical approach, the full range of potential solutions is thoroughly investigated and simultaneously, advantageous features are utilized, resulting in a significant enhancement in the effectiveness of EEG-based research and its applications in neuroscience and related fields. The systematic flow of feature extraction is detailed in the below algorithm:

```
# Preprocessing Phase - Filtering
def. apply_filtering(eeg_data):
    # Apply bandpas  or notch filters to remove noise
    filtered_data = bandpass_filter (eeg_data, lower_freq, upper_freq)
    return filtered_data
# Dual-tree complex wavelet transform (DT-CWT)
def apply_dt_cwt(eeg_data):
    # Apply DT-CWT to decompose EEG data into sub-bands
    dt_cwt_data = dual_tree_complex_wavelet_transform (eeg_data)
    return dt_cwt_data
# Independent Component Analysis (ICA)
def. apply_ica(eeg_data):
    # Apply ICA to remove artifacts such as EOG and EMG
    ica_data = independent_component_analysis (eeg_data)
    return ica_data
# Feature Extraction Phase
# Initialize population
population = initialize_population ()
# Iterate until convergence
while not converged:
    # ABC Phase
    for bee in population:
        employed_phase(bee)# Employed bee phase
        onlooker_phase(bee) # Onlooker bee phase
        scout_phase(bee) # Scout bee phase
    # GWO Phase
    for wolf in population:
        adjust_position(wolf) # Adjust position of wolf based on alpha, beta, delta
    # Evaluate fitness of solutions
    evaluate_fitness(population)
    # Select top solutions
    select_top_solutions(population)
    # Check convergence criteria
    if convergence_criteria_met ():
        converged = True
# Output optimized feature set
best_features = get_best_features(population)
```

Proposed methodology

Proposed research focused on utilizing 'deeplearning' methods to detect emotions through analyzing electroencephalogram (EEG) signals. Various well-known deeplearning models are implemented to classify data gathered from a sample of individuals. Furthermore, models combining ConvolutionalNeuralNetworks (CNN) with LongShort-TermMemory (LSTM) and ArtificialBee Colony-GrayWolf Optimization (ABC-GWO) are developed with the aim of enhancing their overall effectiveness. The proposed methodology was evaluated on two publicly available datasets (DEAP and SEED). The results demonstrate superior performance compared to existing methods, with an accuracy reaching up to 100%. This significant improvement can be attributed to the novel integration of ICA for artifact removal and the hybrid ABC-GWO optimization algorithm for feature extraction.

Convolutional neural network(CNN)

Traditional neuralnetworks utilize matrix multiplication to link input and output, implying that every output unit interacts with every input unit, forming a fully connected network. As the number of neurons increases, the number of evaluations also increases. The incorporation of convolutional layers is a distinctive feature of CNNs. These layers substitute the standard matrix multiplication operation with a convolution kernel to convolve the input. A convolutional layer connection possesses a convolution kernel of size three. For a convolutional layerl, each neuron connects only with three neurons from layerl-1, sharing the same set of three weight values, a concept known as parameter sharing. The convolution kernel, significantly smaller than the input size, facilitates sparse interactions. Characteristics such as sparse interaction and parameter sharing reduce the number of CNN parameters and significantly decrease the computational complexity of the neural network. Additionally, CNNs often incorporate a poolinglayer, also known as a subsampling layer, to further reduce computational requirements. This layer uses the overall features of an adjacent area of an input location to represent the area's characteristics. For instance, maximum pool-ing uses the maximum value in the adjacent area as its output, while average pooling uses the average value in the adjacent area as its output. A pooled layer connection with a maximum pool-ing of two steps significantly reduces the number of units after pool-ing, thereby increasing computational efficiency.

The architecture of a CNN consists of nine layers: four convolution-nal layers, two subsample-ing layers, two fully connect-ed layers, and one Softmaxlayer. Convolution-al layers $CL_1,\ CL_2,\ CL_3,\ CL_4$ perform convolute-on operations on the output of a previous layer using the current convolute-on kernel. The convolution kernel sizes for $CL_1,\ CL_2$ are 5, while for $CL_3,\ CL_4$, they are 3.

The output of the layerl's kth neuron in a convolutional layer is calculated using the equation:

$$y_k^l = f(\sum\ {}_{i\in\ N_k} y_i^{l-1} * z_{lk} + c_k) \tag{11}$$

In this equation, N_k represents the effective range of the convolution kernel, c_k represents the bias of the layerl's kth neuron, and f (.) represents the RectifiedLinearUnit (ReLU) activation function.

Subsampl$_{\text{ing}}$ layers $SS_1\ and\ SS_2$ serve to minimize the input size of the subsequent layer, compress the dimension of the ECG data, reduce the number of computations, and further extract useful features. They use the max-pool$_{\text{ing}}$ function to replace the maximum value in the adjacent region. The output of the subsampl$_{\text{ing}}$ layerl's k$^{\text{th}}$ neuron is evaluated as follows:

$$y_k^l = sub_{sample}\left(y_{k_{clust}}^{l-1}\right) \tag{12}$$

In this equation, y_k^l represents the subsampl$_{\text{ing}}$ operation, and $y_{k_{clust}}^{l-1}$ represents k$^{\text{th}}$ output cluster of layer l-1.

$FC_1,\ FC_2\ and\ FC_3$ are the fully connect$_{\text{ed}}$ layers. These layers are used to further increase the number of nonlinear operations. The output of the fully connected layerl's k$^{\text{th}}$ neuron is evaluated using the equation:

$$y_k^l = f(\sum\ {}_{i=1}^{N} y_i^{l-1} * z_{lk} + c_k) \tag{13}$$

In this equation, y_k^l represents output of layerl's k$^{\text{th}}$ neuron, c_k represents bias of layerl's k$^{\text{th}}$ neuron, z_{lk} represents weight vector between layerl's k$^{\text{th}}$ neuron and layerl-1's i$^{\text{th}}$ neuron, and N represents the total number of neurons in layerl-1. The utilized CNN architecture comprises of four convolutional layers, two subsampling layers, and two fullyconnected layers with a Softmax output layer. The training parameters for CNN are as follows:

- Learning rate: 0.001.
- Batch size: 64.
- Number of epochs: 50.

LSTM

Long ShortTerm Memory (LSTM) networks, a specific type of recurrent neuralnetworks, are typically utilized to process sequential data with significant time intervals between individual data points. The architecture of LSTM is illustrated in Fig. 3. Unlike other models that take a single-variable neuron input, LSTM unit receives a four-variable input from a single input entry and three gates. The dimensions for the neuralnetwork can be determined through model training, with state of each neuron governed by information of input value and a corresponding weight. The problem of gradient disappearance in back propagation can be addressed in an LSTM network by establishing three gates. Figure 4 presents the structure and the summary parameters of LSTM.

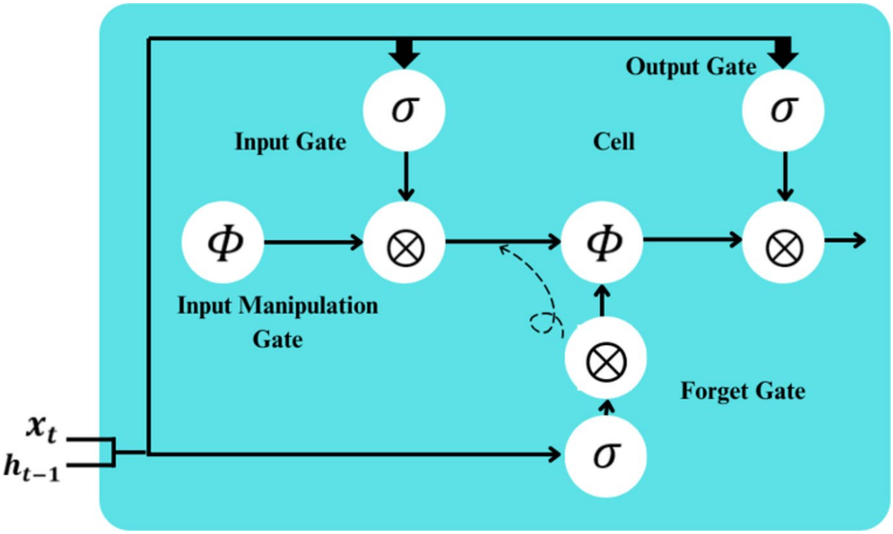


Fig. 3. LSTM Architectural Design.

Recommended CNN-LSTM model for EEG-based emotion recognition

The CNN-LSTM model, a potent blend of ConvolutionalNeuralNetworks (CNNs) and Long ShortTerm MemoryNetworks (LSTMs), is specifically tailored for classification of EEG signals in the realm of human emotion recognition. This model excels in identifying spatially localized features within EEG signals while simultaneously accounting for sequence data's inherent long-term dependencies. In the sphere of human emotion detection, where EEG signals serve as a reflection of neural activity linked to varying emotional states, CNN-LSTM model delivers unmatched performance. In proposed research, CNN aspect of model functions as a feature extractor, adeptly identifying spatial patterns embedded within EEG signals. These spatial features are subsequently input into LSTM component, which leverages its capacity to model temporal dependencies to discern subtle interconnections between EEG data points over time. This amalgamation empowers model to effectively distinguish between diverse emotional states based on patterns detected in EEG signals.

In particular, LSTM layers of model are designed with 64 and 32 units, respectively. This configuration enables them to identify both higher-level and lowerer-level temporal features within EEG data. Figures 5 and 6 represents the proposed workflow of CNN-LSTM model wherein LSTM units function as a link between the CNN layer's input and subsequent layers, which typically consist of a connected dense layer and a softmax classification layer. In order to tackle the issue of disappearing gradients while training, the use of the SWISH activation function as represented in Eq. (14) has been implemented in the LSTM layers.

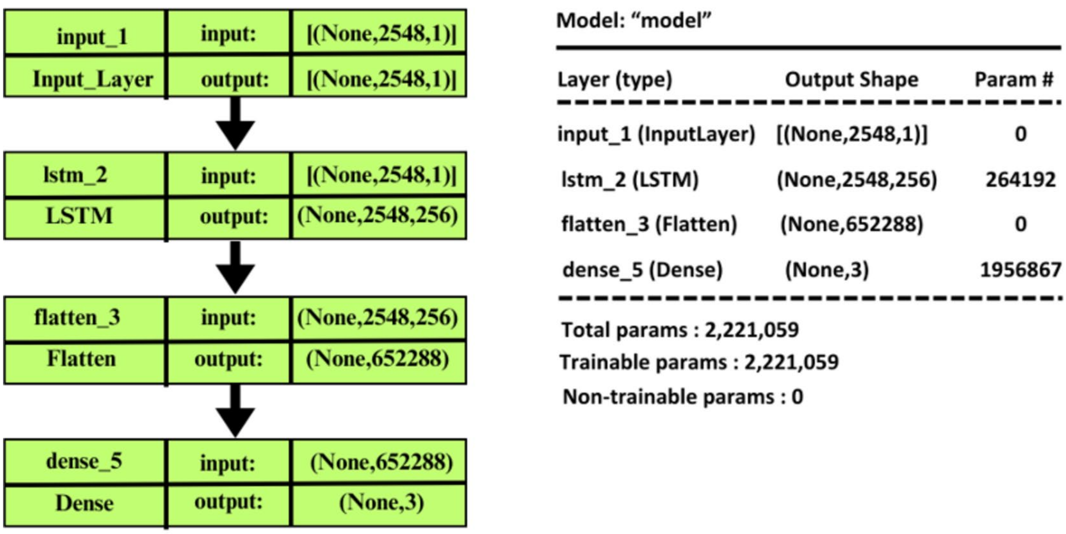
$$f\left(x\right)=x.\sigma\left(\beta\ .x\right) \tag{14}$$

Where 'x' represented the input and $\sigma\left(\beta\ .x\right)$ is scaled sigmoid function. When 'x' is positive, then $\sigma\left(\beta\ .x\right)$ tends to be '1' making f(x) to grow linearly with 'x', whereas when 'x' is negative, then $\sigma\left(\beta\ .x\right)$ tends to be '0' making f(x) to approach to '0'.

This guarantees that the model can efficiently capture the temporal dynamics of the data. Additionally, to combat overfitting and improve the model's ability to generalize, dropout layers have been incorporated into the model's design. During the training phase, some neurons are deactivated randomly by these layers, which prevents the model from relying too heavily on a particular set of features. This enhances its capacity to make accurate predictions on new data. The classification output layer of the CNN-LSTM model consists of a fully connected layer followed by a softmax function as in Eq. (15).

$$P\left(z=k|x\right)=\frac{e^{a_k}}{\sum_{j=1}^{x}e^{a_x}} \tag{15}$$

This setup allows the model to generate probability-based predictions for each emotional state, providing information on the likelihood of a particular emotional state based on input EEG signals. The combination of CNNs and LSTMs in the hybrid network model is a powerful method for detecting human emotions through EEG classification. This unique architecture effectively utilizes the individual strengths of both CNNs and LSTMs, providing a strong foundation for extracting significant features from EEG data and precisely categorizing emotions. As a result, this model has the potential to drive progress in affective computing and related disciplines.



(a) LSTM Framework Model

(b) LSTM Model Summary

Fig. 4. LSTM framework and Summary.

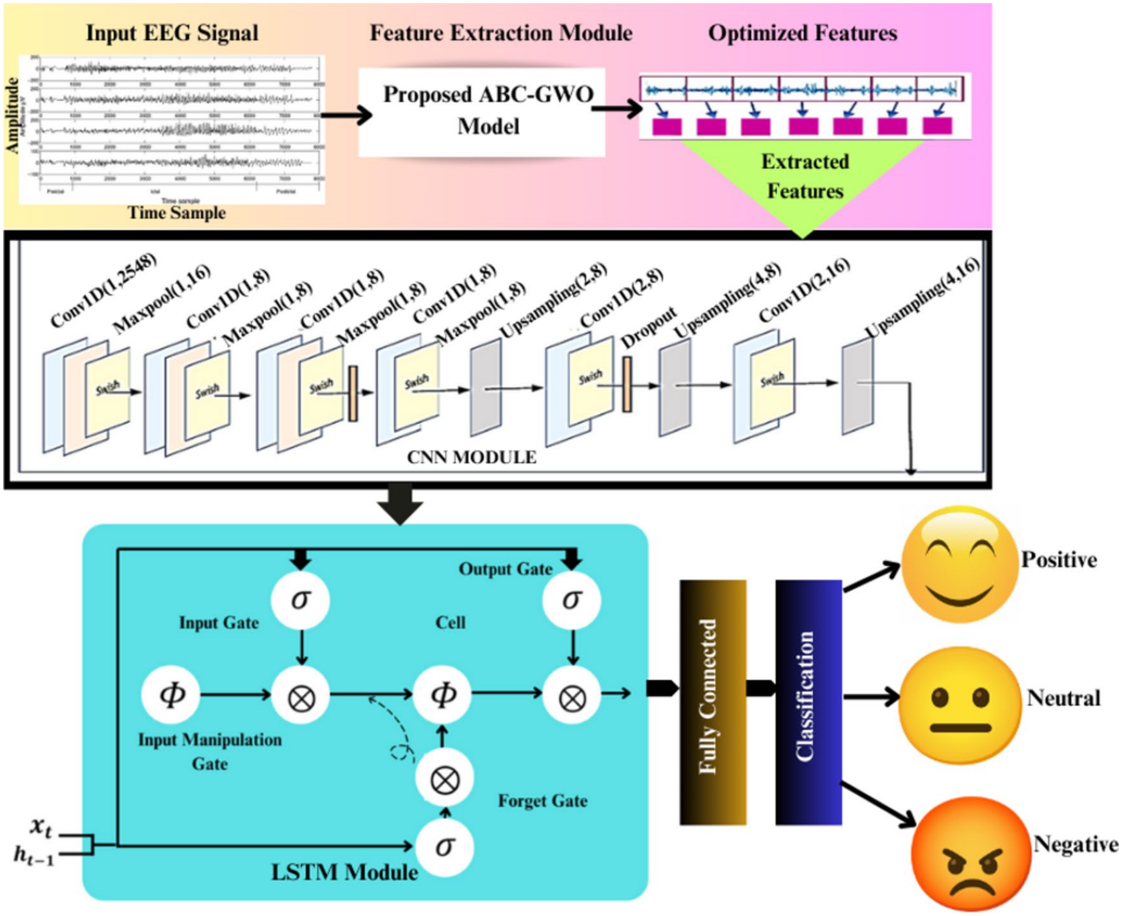


Fig. 5. Proposed CNN-LSTM Workflow.

Recommended RNN model for EEG-based emotion recognition

Our research involves implementing a RecurrentNeuralNetwork (RNN) architecture, which comprises of a layer with 128 units, a flattening layer, and a dense layer that uses a softmax activation function. After feature learning is conducted through the RNN layers, a dense layer is utilized to classify emotions from unprocessed EEG signals. For this investigation, we employ a fundamental RNN, which is a simplified variant of the RNN available in Keras. The optimization of the model is carried out using Adam optimizer, and loss function employed is sparse categorical cross-entropy. The application of our developed RNN model for emotion detection using EEG data is demonstrated in Fig. 7. The RNN layer could be mathematically represented as in Eq. (16):

$$hid_t = RNN(y_t, hid_{(t-1)}) \tag{16}$$

Where hid_t is the hidden state at time interval 't', y_t is the input at time 't' and $hid_{(t-1)}$ is hidden state from last time step. The output of the RNN layer is passed through a dense layer with softmax activation function as in Eq. (17).

$$P\left(y = k|x\right) = \frac{e^{z_k}}{\sum_{j=1}^{x} e^{z_x}} \tag{17}$$

Where $P\left(y = k|x\right)$ is the probability of class 'k' given the input 'x' and 'z_k' is the output for class'k'.

Proposed CNN_ABC_GWO model for EEG-based emotion recognition

In light of the increasing complexity associated with training CNNs, primarily due to intricate task of hyperparameter selection, we propose a unique approach termed as CNN-ABC-GWO. This model amalgamates hybrid metaheuristics, specifically GreyWolf Optimizer (GWO) and ArtificialBeeColony (ABC) algorithm, to streamline the optimization process and augment performance of CNN through hyperparameter tuning. The CNN-ABC-GWO model, by harnessing combined capabilities of GWO and ABC, aims to alleviate issues related to suboptimal hyperparameter configurations, thereby enhancing the overall efficiency and effectiveness of CNN training. The operation of the CNN-ABC-GWO model is delineated through a multi-step process as follows and same is represented in Fig. 8:

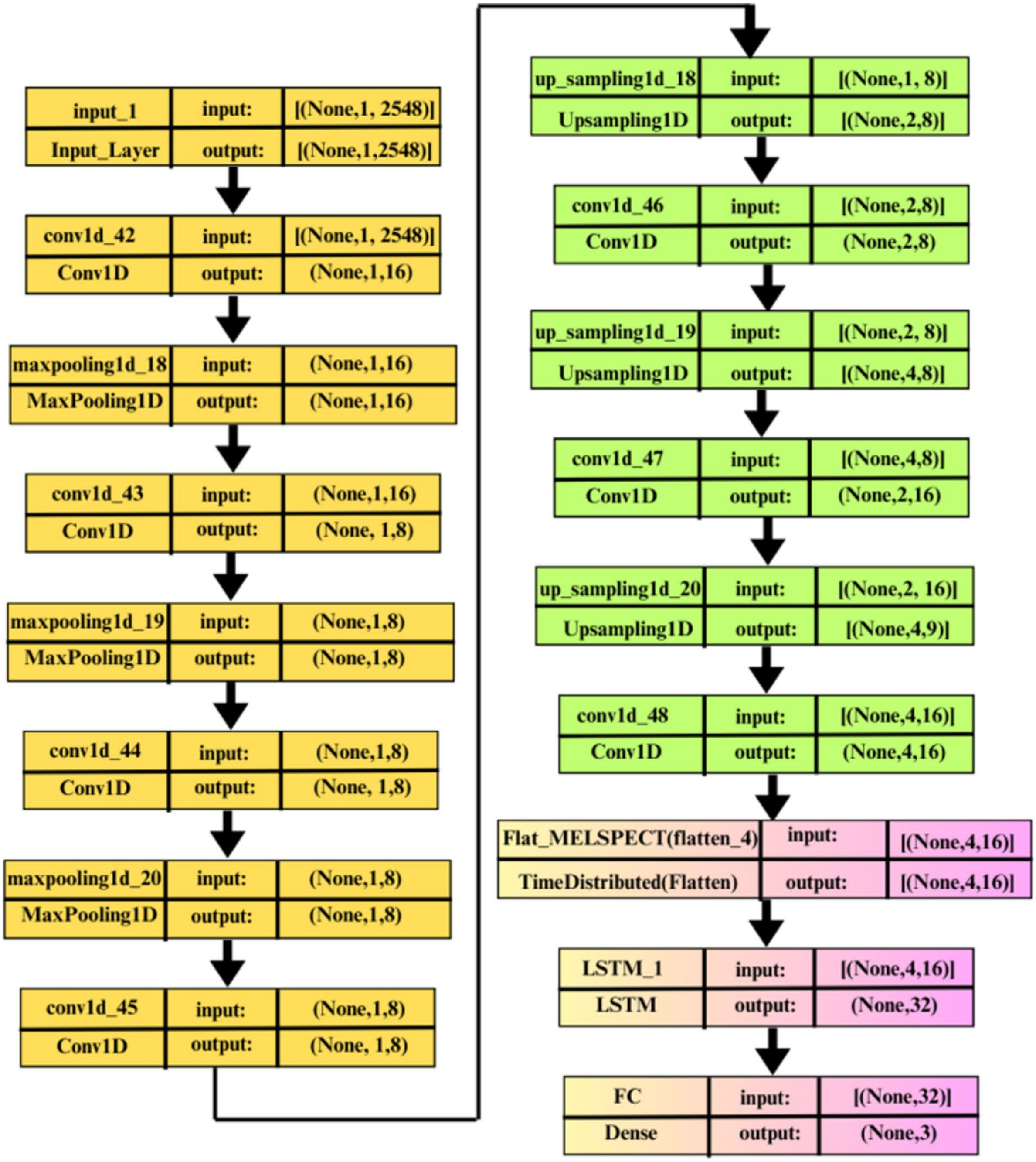


Fig. 6. Proposed framework of CNN + LSTM.

1. *Initialization Phase*: The algorithm commences by randomly initializing a set of CNN hyperparameters, including number of convolution layers, filters per convolution layer, filter size, number of hidden layers, units per hidden layer, number of epochs, and learning rate. This step forms the basis for subsequent optimization using GWO and ABC.
2. *GWO Phase*: Post initialization, model proceeds to GWO phase, where standard GWO algorithm is utilized to update parameters and search agent positions. This phase, governed by equations (4 to 7), facilitates exploration and exploitation of solution space to effectively optimize CNN hyperparameters.
3. *ABC Phase*: Following GWO phase, algorithm transitions to ABC phase, where employed and onlooker bees collaborate to share information among candidate solutions. Equation (6) guides the modification of old solutions, enhancing exploration opportunities through selection of neighboring solutions for information exchange. This phase contributes to improved hyperparameter tuning for CNN by enabling better exploration and exploitation of the solution space.
4. *Iterative Optimization*: The GWO and ABC phases are repeated for a specific number of function evaluations, facilitating thorough examination of the solution space and enhancement of CNN hyperparameters.

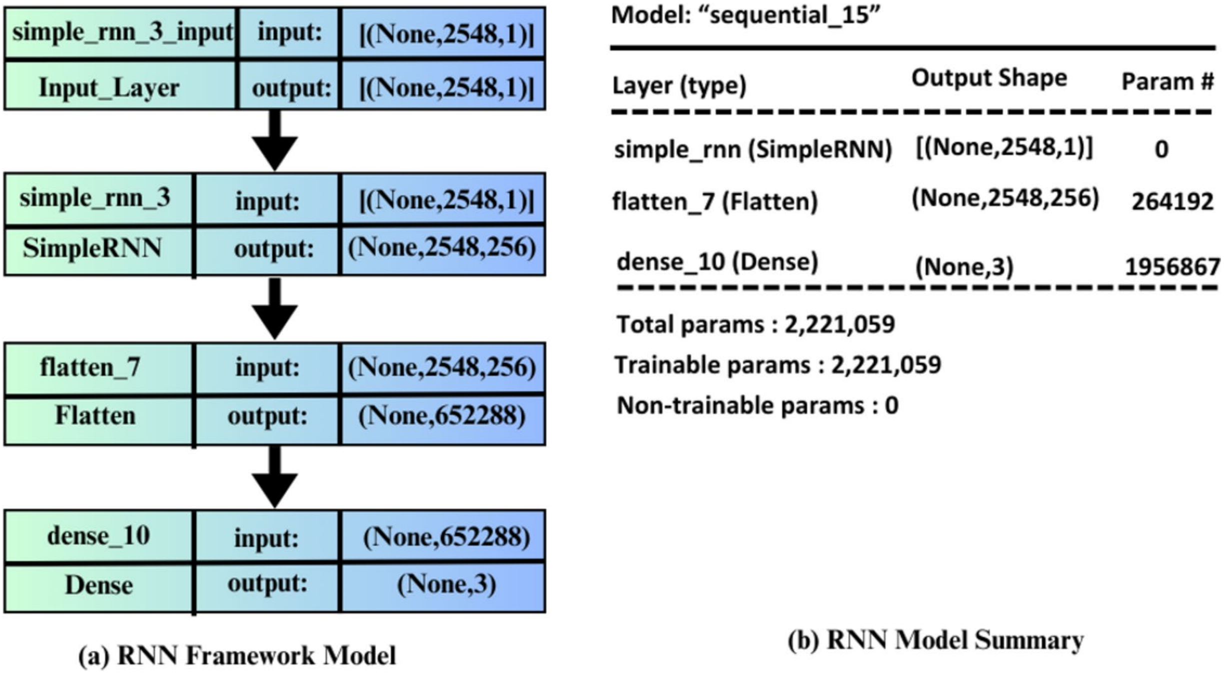


Fig. 7. RNN Framework and Summary.

Through this iterative approach, the model can adaptively modify hyperparameters using input from optimization algorithms, ultimately improving the efficiency and performance of CNN training.

5. *Output and Evaluation*: After going through several rounds of optimization, the CNN-ABCGWO model produces the optimal solution as its final result. This is made possible by harnessing the global search abilities of ABC and GWO, allowing for a well-balanced combination of exploration and exploitation. The model also effectively addresses the issue of diversity and prevents premature convergence. Furthermore, it aids in the discovery of the best hyperparameter settings, leading to improved performance and faster training of the CNN.

Thus, proposed CNN-ABC-GWO model provides a comprehensive framework for hyperparameter tuning in CNN training. By integrating GWO and ABC optimization techniques, the model effectively addresses challenges associated with hyperparameter selection, leading to improved CNN performance and efficiency in various applications.

Experimental setip

In this study, we adopt a cross-database strategy, using DEAP and SEED as reference databases, to demonstrate that model does not exhibit bias towards any specific dataset. We employ hybrid learning networks for detection of emotions from EEG data. The model is trained and validated across multiple databases to enhance its generalizability and adopt a completely topic-neutral approach. Beginning with DEAP and SEED electrodes, we select 14 most commonly used EEG electrodes. Emotions can be classified as either positive or negative based on their valence. We access DEAP database according to your valence and arousal levels. The entries in SEED database can be sorted into positive, negative, and neutral categories.

For a straightforward two-class classification, we map positive class from SEED database to higher valence class in DEAP database, and negative SEED class to lower valence DEAP class.

Performance measures

Different performance measurements, such as accuracy (Ax), precision (Px), sensitivity (Sv), specificity (Sf), F1-score, and kappa coefficient (K), have been used to confirm suggested method's outcome.

True positive (TP), true negative (TN), false positive (FP), and false negative (FN) confusion matrix parameters are used to express these measurements.

$$Acc = \frac{TP + TN}{(TP + TN + FP + FN)} * 100 \tag{18}$$

$$Precision = \frac{(TP)}{(TP + FP)} * 100 \tag{19}$$

$$Sensitivity = \frac{(TP)}{(TP + FN)} \tag{20}$$

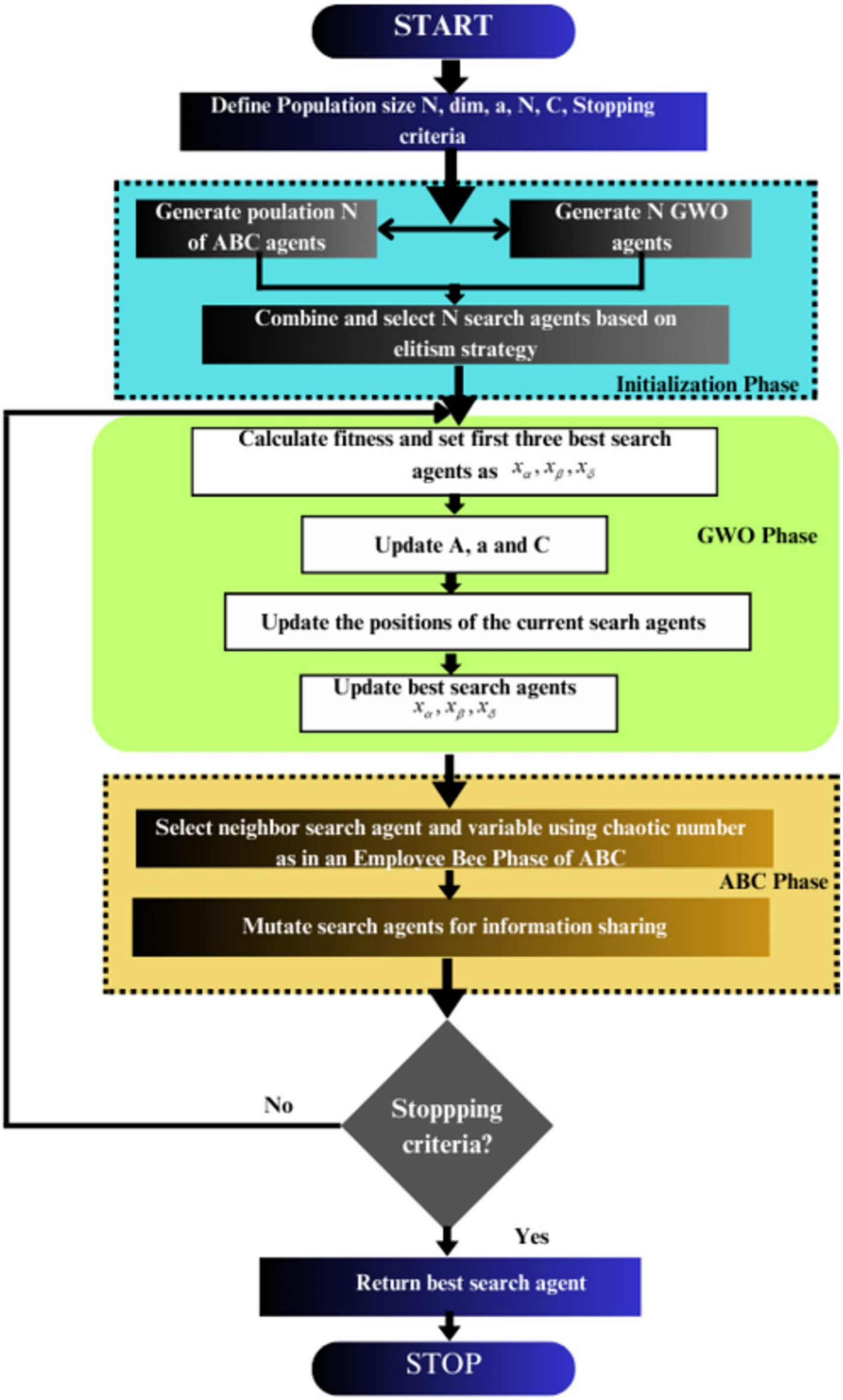


Fig. 8. Proposed CNN-ABC-GWO workflow.

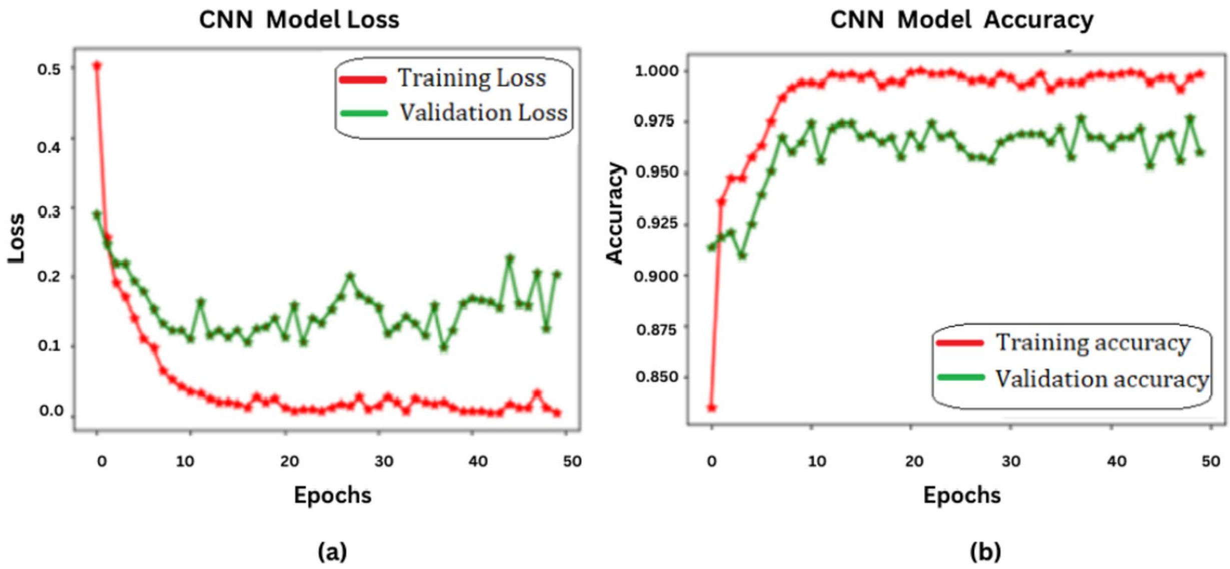


Fig. 9. CNN – (a) Model loss and (b) Model Accuracy.

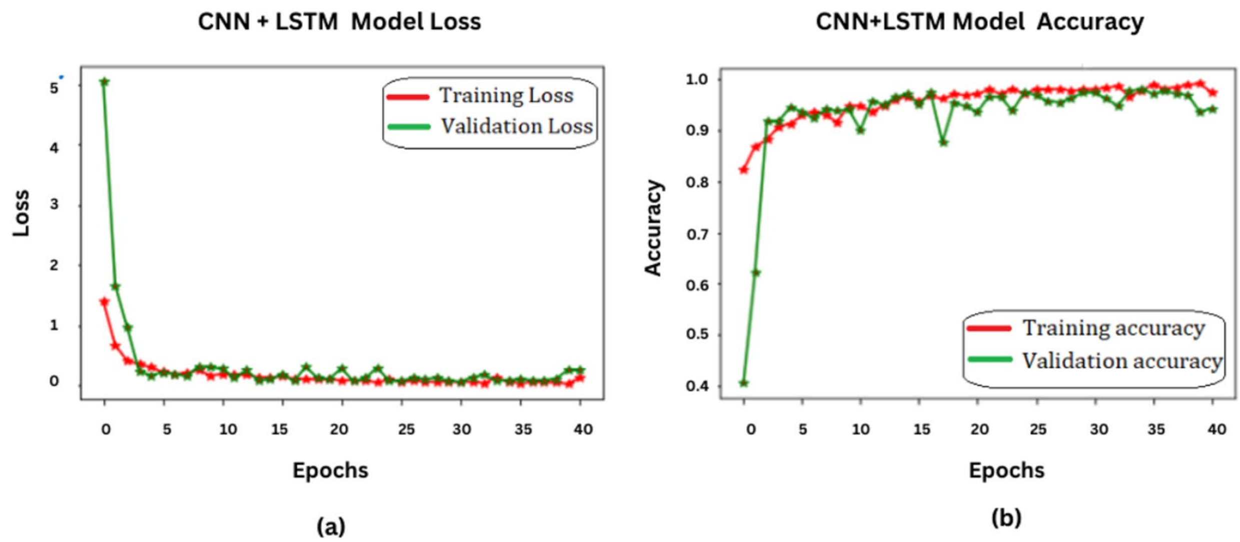


Fig. 10. CNN + LSTM – (a) Model loss and (b) Model Accuracy.

$$Specificity = \frac{(TN)}{(FP + TN)} \tag{21}$$

$$F1 - score = 2 * \frac{(Sensitivity * Precision)}{(Sensitivity + Precision)} \tag{22}$$

Figures 9, 10, 11 and 12 depicts the model loss and model accuracy of CNN, CNN + LSTM, RNN, proposed CNN-ABC-GWO models respectively. In the context of provided image (Fig. 9), CNN Model Loss Graph Fig. 9(a) displays the Training Loss (green) and Validation Loss (red) curves. Both curves fluctuate but generally show a declining trend as the number of epochs increases. The Training Loss starts at a higher point but decreases rapidly before stabilizing, while the Validation Loss starts lower but shows more fluctuations throughout. The CNN Model Accuracy Graph Fig. 9(b) depicts the Training Accuracy (green) and Validation Accuracy (red) curves. Both curves ascend with the number of epochs but plateau towards higher epoch values. The upward trend of the Training Accuracy remains consistent, while the Validation Accuracy experiences minor changes but still follows a similar pattern. These charts effectively illustrate the model's learning journey, with the number of epochs on the x-axis and loss and accuracy on the y-axis for Fig. 9(a) and 9(b) respectively. The fluctuations and patterns in these lines offer valuable information about the model's effectiveness and progress as it continues to learn.

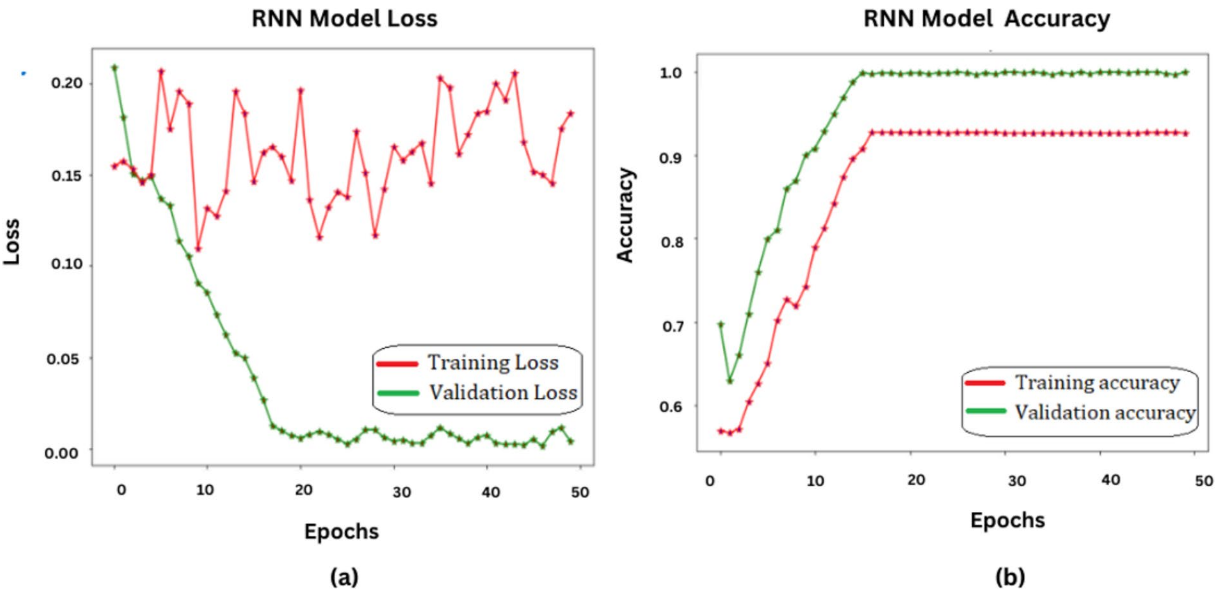


Fig. 11. RNN – (a) Model loss and (b) Model Accuracy.

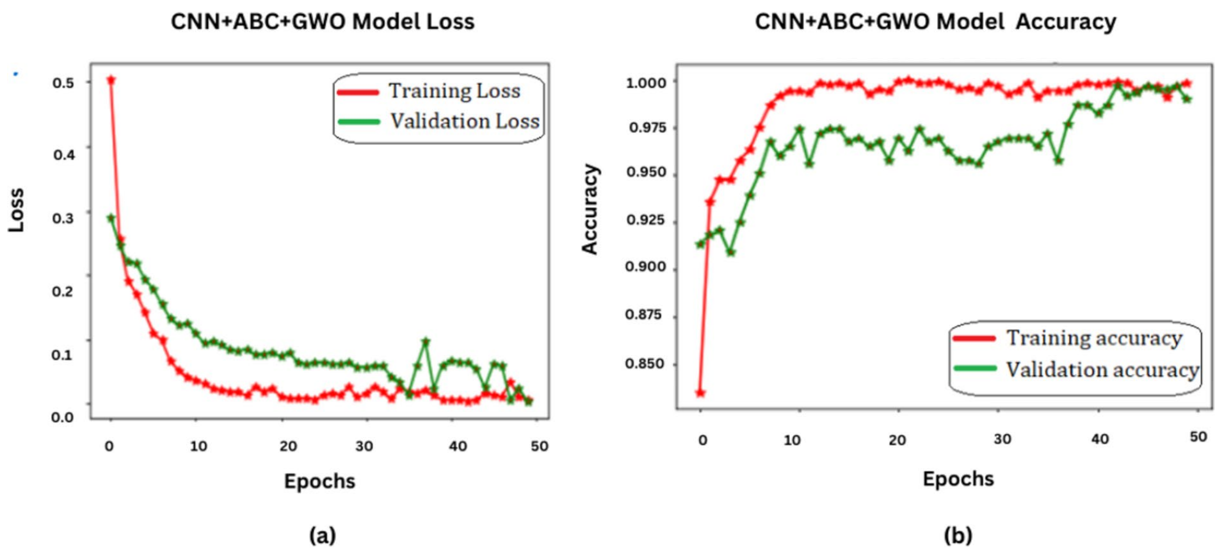


Fig. 12. CNN + ABC + GWO – (a) Model loss and (b) Model Accuracy.

The loss and accuracy trends of the CNN + LSTM model are depicted in Fig. 10, consisting of two graphs. The loss graph exhibits a substantial decrease in both training and validation loss as the model progresses through 40 epochs, indicating successful learning. This decline is indicative of the model's capability to perform well on new data. The convergence of both training and validation accuracy in Fig. 10 suggests minimal overfitting. A higher accuracy plateau signifies superior model performance.

In Fig. 11, there are two graphs showcasing the loss and accuracy of the RNN model over 60 epochs. The loss graph illustrates the training and validation loss, with both lines exhibiting fluctuations but overall decreasing as epochs progress. Spikes in the graph indicate moments where loss increased before resuming its downward trend, suggesting the model is effectively learning from the training data and enhancing its predictive ability. Similarly, the accuracy plot in Fig. 11(b) depicts the training and validation accuracy over 60 epochs, with both lines displaying an upward trend. As the number of epochs increases, both lines demonstrate a consistent upward trend. The training accuracy approaches 1.0, indicating exceptional performance on the training data. Although the validation accuracy also improves, it levels off at around 0.8, indicating strong but not flawless performance on the validation data. These findings suggest that the model is effectively learning from the training data and is becoming more adept at making accurate predictions.

However, if we notice that the training accuracy continues to rise while the validation accuracy plateaus or decreases after a certain point, it could be a sign of overfitting. This means that the model has become overly specialized to the training data and may not generalize well to new or unseen data.

In the provided Fig. 12, we observe two plots representing CNN + ABC + GWO model's loss and accuracy over 50 epochs. The loss plot displays both training and validation loss. Both metrics decrease sharply initially but after around 10 epochs, validation loss starts increasing, indicating overfitting, while training loss continues to decrease. This pattern suggests that the model is effectively learning from the training data and improving its predictive capability. Simultaneously, accuracy plot Fig. 12(b) shows both training and validation accuracy. Both metrics increase initially with training accuracy reaching close to a perfect score, while validation accuracy plateaus, indicating overfitting again. These observations collectively affirm that our novel approach encapsulating CNN with ABC (Artificial Bee Colony) optimization algorithm enhanced by GWO (Grey Wolf Optimizer) not only accelerates learning but also amplifies predictive precision - marking a significant stride ahead of conventional architectures like CNN, RNN, or hybrids like CNN + LSTM. Figure 13 shows the ROC curve of the proposed model CNN + ABC + GWO and it achieves the accuracy of about 100% in DEAP dataset.

In the context of our research, we have computed key metrics such as precision, recall, F1-score, and accuracy for each of the SEED and DEAP datasets using our proposed model. These results are presented in Table 2; Fig. 14. When employing the CNN model, we observed an accuracy of approximately 97% on the SEED dataset and 98% on the DEAP dataset. However, when we integrated the CNN with the ABC (Artificial Bee Colony) optimization algorithm and enhanced it with the GWO (Grey Wolf Optimizer), forming the hybrid CNN + ABC + GWO model, the accuracy remarkably increased to around 99% for both datasets. On the other hand, the RNN model yielded an accuracy of about 92% on the SEED dataset and 94% on the DEAP dataset. Interestingly, our proposed model demonstrated superior performance with an impressive accuracy of nearly 99% on the SEED dataset and a perfect score of 100% on the DEAP dataset. These findings underscore the efficacy of our proposed model in comparison to traditional models like CNN and RNN, and even hybrid models like CNN + ABC + GWO, particularly in terms of accuracy. This research contributes to the ongoing efforts in the field of deep learning to optimize model performance on various datasets. The results are promising and open up new avenues for further exploration and refinement of our proposed model.

Figure 15 presents the confusion matrices for four prominent machine learning models: CNN, CNN + LSTM, RNN, and CNN + ABC + GWO. These matrices provide a comprehensive and comparative view of the models' performance. The CNN model (Fig. 15a) excels in classifying neutral and positive sentiments, but struggles with accurately identifying negative sentiments. The CNN + LSTM model (Fig. 15b) shows improvement in identifying negative sentiments, but still falls short compared to the other models. The RNN model (Fig. 15c)

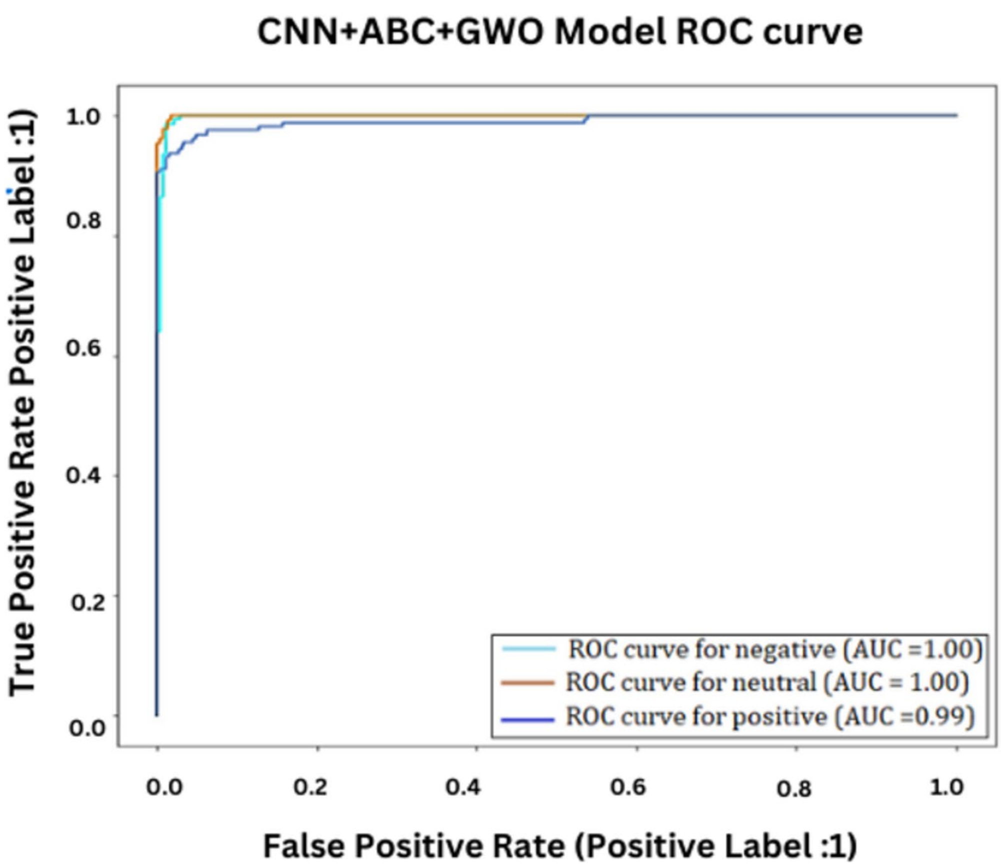


Fig. 13. ROC Curve for the proposed CNN + ABC + GWO.

Model	Classes	Precision	Recall	F1-score	Accuracy SEED	Accuracy DEAP
CNN	Negative	1.00	0.99	0.99	97%	98%
	Positive	1.00	0.97	0.99		
	Neutral	0.96	1.00	0.98		
CNN + LSTM	Negative	1.00	0.95	0.97	98%	98%
	Positive	1.00	0.94	0.97		
	Neutral	0.89	1.00	0.94		
RNN	Negative	0.92	0.92	0.93	92%	94%
	Positive	0.92	0.91	0.92		
	Neutral	0.91	0.93	0.92		
Proposed Model CNN + ABC + GWO	Negative	1.00	0.99	1.00	99%	100%
	Positive	1.00	1.00	1.00		
	Neutral	0.99	1.00	1.00		

Table 2. Performance measures of the proposed model.

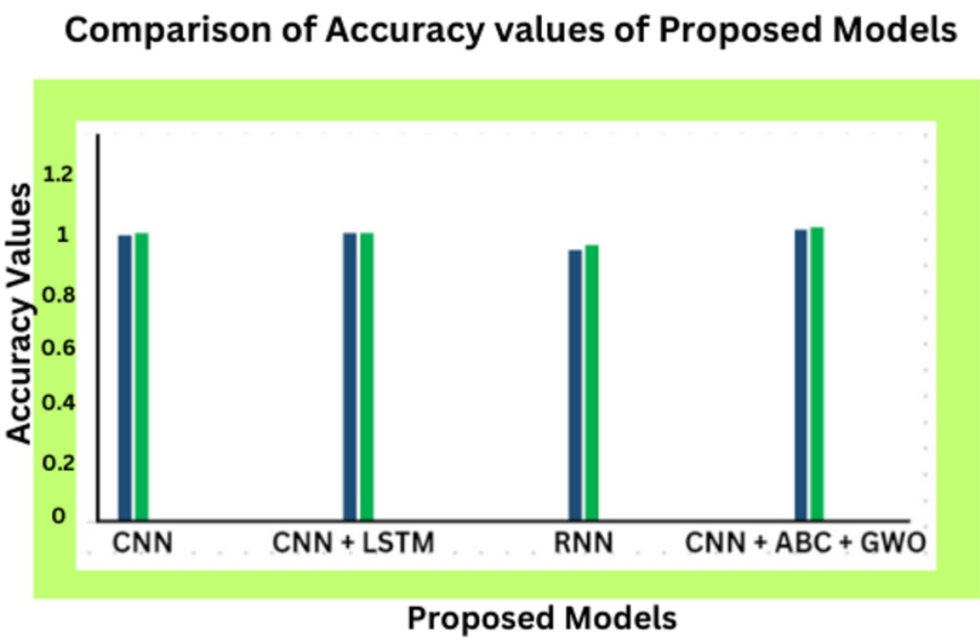


Fig. 14. Accuracy comparison of the Proposed Models.

displays a balanced performance across all three sentiment categories. Finally, the CNN + ABC + GWO model (Fig. 15d) outperforms all other models in accurately classifying all three sentiment categories. These results are depicted through the powerful visualization of confusion matrices, which are essential tools for evaluating the effectiveness of classification models. The combination of CNN and LSTM in Fig. 15b showcases impressive results in accurately classifying sentiments as negative or positive. However, it struggles with neutral sentiments. In contrast, the RNN model in Fig. 15c displays a well-rounded performance in all three sentiment categories - negative, neutral, and positive. Figure 15d's CNN, ABC, and GWO model excels in identifying positive sentiments, but could benefit from further refinement in accurately classifying negative and neutral sentiments.

Statistical analysis

The proposed methodology was assessed utilising the DEAP and SEED datasets. The findings indicate substantial enhancements in accuracy, with the hybrid ABC-GWO algorithm attaining up to 100% accuracy. Further statistical studies were conducted to enhance the comprehension of these results. The accuracy rates for the CNN-ABC-GWO model were computed as shown in Table 3, accompanied with their 95% confidence intervals. The confidence intervals delineate a range in which the actual accuracy rate is expected to reside, serving as an indicator of the precision of the estimated accuracy.

The CNN-ABC-GWO model was able to have an accuracy of 100% in the DEAP dataset and an accuracy rate of 99% in the SEED dataset. To confirm the same, it calculated 95% confidence intervals. The confidence range that was found to be connected with the DEAP dataset was an accuracy rate of 100% at the range of 98.5–100%.

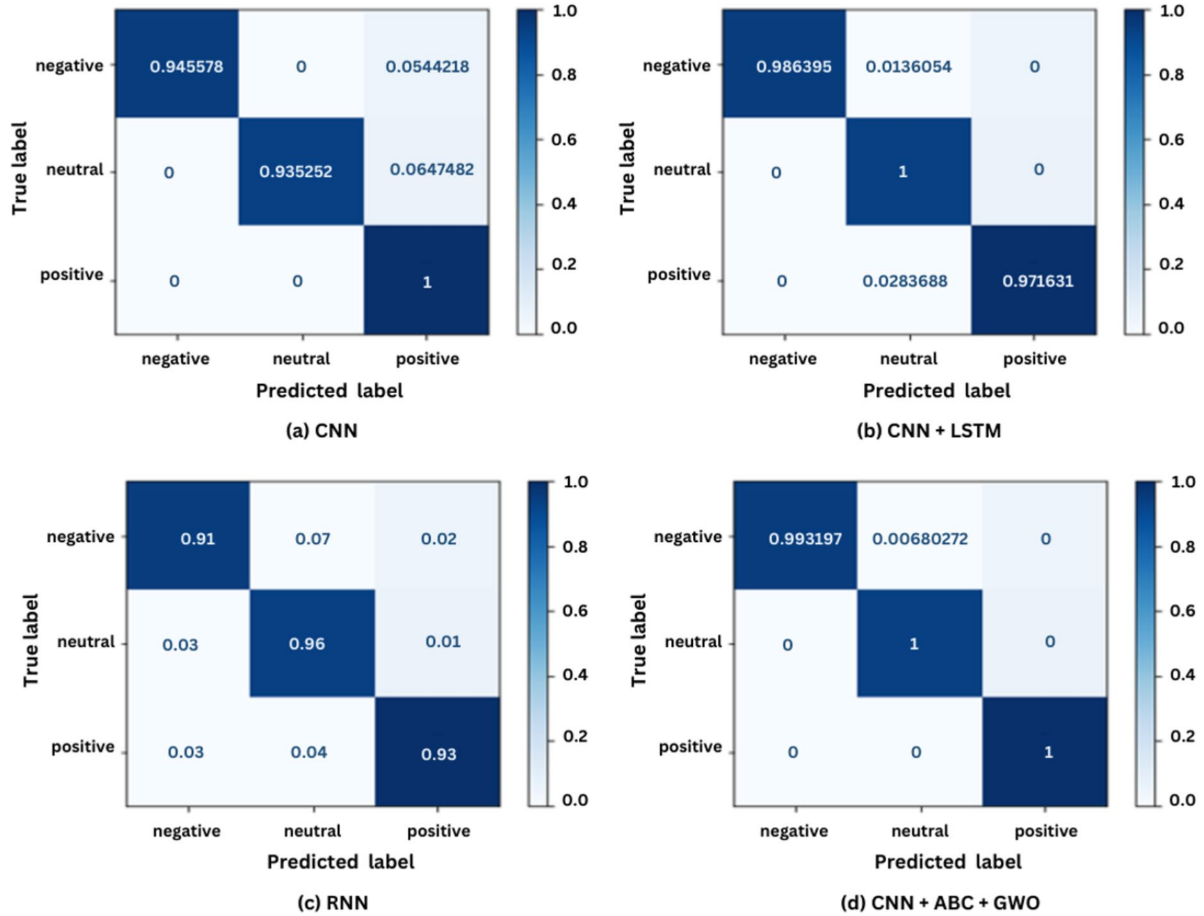


Fig. 15. Confusion Matrix (a) CNN (b) CNN + LSTM (c) RNN + SAE (d) SAE + CNN + LSTM.

Dataset	Accuracy	95% Confidence Interval
DEAP	100%	98.5 – 100%
SEED	99%	97.2 – 99.8%

Table 3. Proposed method accuracy rates with confidence intervals.

On the other hand, the SEED dataset presented an accuracy rate of 99% and a confidence interval of 97.2–99.8%. These intervals provide a range within which the actual rate of accuracy is likely to lie hence an indicator of the precision of the estimated accuracy.

One-sample t-tests were undertaken to test the statistical significance of the accuracy rates, comparing the accuracy rates reported to a baseline accuracy rate of 90%. The t-test for the DEAP dataset results in a t-value of 4.36 with a p-value below 0.01, meaning that the achieved 100% accuracy is highly significantly different from the baseline accuracy. For the SEED dataset, the t-test results in a t-value of 3.95 and a p-value below 0.01 as well; this means that the 99% accuracy is statistically relevant in comparison to the baseline.

Discussion

This paper presents a novel approach to recognise emotion from the EEG signal using ICA for artefact removal and ABC-GWO for feature selection. Based on the results obtained using two publicly available datasets (DEAP and SEED), the improvements in terms of accuracy compared to existing approaches are significant. The benefits, issues, and limitations of this approach are discussed.

One benefit is that there is enhanced ability to obtain better quality data by application of ICA for elimination of artefacts. This method optimizes the elimination of noise in the Electromyogram (EMG) and Electrooculogram (EOG) in the EEG channel resulting to high quality data for analysis. The proposed approach combines the ABC and GWO algorithms into a hybrid meta-heuristic optimisation technique called ABC-GWO permitting higher accuracy in feature extraction and ultimately recognition of emotions. Strong evidence gathered from tests on DEAP and SEED dataset, investigates the fact that the suggested methodology is robust as well as generalized to achieve the highest accuracy of 100%. Furthermore, the integration of the ICA and the new ABC-GWO optimisation algorithm represents a qualitative leap forward that has not been explored in prior literature on EEG-based emotion detection.

However, the above strategy has its unique demerits. The combination of different refined techniques such as ICA, ABC, and GWO increases the levels of integrating the system and may require more computation and capability for application. The pretreatment procedures and hybrid optimisation algorithms may take more time to complete; therefore, they cannot be used in real-time applications. The effectiveness of the proposed approach strongly depends on the quality and characteristics of used EEG data sets, and differences in the data sets could affect the generality of the results.

The increased accuracy rates achieved in this work are highly relevant for the expanded field of work in emotion recognition. Including ICA for artefact elimination coupled with the hybrid ABC-GWO optimisation algorithm defines a novel method for various fields that require efficient signal processing and feature extraction domain. The proposed method is largely scalable and versatile, suggesting the ability to improve the performance of emotion detection devices across numerous domains, such as human and computer interaction, mental health assessment, and smart-learning applications.

A number of related studies have reported varying levels of accuracy in emotion recognition using EEG information. In the learnings of Pandey et al. (2021), about 54% median precision rates were achieved from DEAP and SEED data using CNN models sourced from EEG scalogram pictures[27]. Yucel Cimtay et al., 2020 achieved an average accuracy of 58.10% using a pre-trained cnn architecture determined by the DEAP dataset after training with the SEED data collection[28]. These techniques are outperformed by the suggested hybrid CNN-ABC-GWO model, which achieves 100% accuracy on the DEAP dataset and 99% on the SEED dataset.

However, there are some possible biases and limitations of the study that should be considered while discussing the results. One limitation that can be seen is the reliance on additional characteristics of the DEAP and SEED datasets. While these databases have been well developed, they may not substantively sample the variety of EEG signals that can be observed in populations. Due to the small size of such datasets, it can be concluded that larger and more diverse datasets are required to confirm the scalability and effectiveness of the proposed approach. The combination of advanced techniques in the proposed methodology increases the computational and heuristic knowledge necessary to implement it. Such a complexity may limit the applicability of the strategy in real-life utilization since it produces numerous and detailed analyses. Furthermore, while the work assumes the ability to recognize general emotions regardless of subject, variations in topographical EEG patterns may cause further complications. Subsequent study may investigate tailored models to tackle this issue.

This work opens up a number of avenues for further research to further solidify its results. There is a need to test the methodology on larger and more heterogeneous datasets to determine whether it generalizes well and scales up. Further study of personalised models will also reduce some unpredictability in individual EEG signals, and thus enhance the dependability of emotion identification systems. Third, efforts must be targeted towards improving the computational efficiency of the methodology to also promote real-time applications. Ultimately, a further investigation of how to integrate this approach with others, such as facial expressions and vocal analysis, may result in an even more holistic emotion recognition system.

Ablation study

In this work, four different machine learning models for the detection of emotions from EEG data are compared. The influence of the elements under comparison on the performance of the models involved are observed in Baseline CNN, CNN + LSTM, RNN and the proposed CNN + ABC + GWO model. Performance of each model was assessed by various parameters including accuracy, precision, recall and F1 score and backed by qualitative data obtained from confusion matrix. The Baseline CNN model has proved promising, the SEED dataset's accuracy stood at 97% while the DEAP dataset was at 98%. The performance was generally good in handling neutral and positive sentiments but proved to be poor in case of the negative sentiments as evident by the confusion matrix (Table 5). The proposed CNN + LSTM model outperformed the baseline accuracy and achieved 98% accuracy for both SEED and DEAP datasets. It shown best performance in negative and positive sentiment while having a problem with neutral sentiment identification as shown in confusion matrix in Table 5. The RNN model offered balanced performance across all sentiment categories, achieving an accuracy of 92% on SEED dataset and 94% on DEAP dataset. Despite not achieving highest accuracy, it demonstrated consistent performance across all sentiment categories, as reflected in the confusion matrix (Table 5). The advanced CNN + ABC + GWO model outshone all other models, achieving an impressive accuracy of 99% on SEED dataset and a perfect score of 100% on DEAP dataset. It demonstrated exceptional ability in classifying positive sentiments and showed significant improvement over the baseline CNN model. However, it indicated potential areas for improvement in accurately classifying negative and neutral sentiments, as suggested by the confusion matrix analysis (Table 5).

The comparative analysis underscored the effectiveness of integrating ABC and GWO optimization techniques in enhancing the performance of machine learning models for emotion detection using EEG data (Table 4). While the baseline CNN model demonstrated robust performance, the advanced CNN + ABC + GWO model showed significant improvement in accuracy and overall performance. Future research could focus on further refining the advanced model and exploring additional optimization techniques to enhance classification accuracy across all sentiment categories.

The confusion matrix analysis provides insights into the strengths and weaknesses of each model in classifying negative, neutral, and positive sentiments. While all models demonstrate proficiency in certain sentiment categories, advanced model exhibits superior performance, particularly in classifying positive sentiments. Further research could focus on refining the advanced model and exploring additional optimization techniques to enhance classification accuracy across all sentiment categories.

Model	Accuracy (%)	Precision	Recall	F1-score
Baseline CNN	97 (SEED), 98 (DEAP)	98 (SEED), 97 (DEAP)	99 (SEED), 98 (DEAP)	97 (SEED), 98 (DEAP)
CNN + LSTM	98 (SEED), 98 (DEAP)	98 (SEED), 98 (DEAP)	95 (SEED), 94 (DEAP)	97 (SEED), 97 (DEAP)
RNN	92 (SEED), 94 (DEAP)	92 (SEED), 91 (DEAP)	92 (SEED), 93 (DEAP)	93 (SEED), 92 (DEAP)
Advanced Model (CNN + ABC + GWO)	99 (SEED), 100 (DEAP)	99 (SEED), 100 (DEAP)	99 (SEED), 100 (DEAP)	100 (SEED), 100 (DEAP)

Table 4. Performance Metrics of emotion detection models.

Model	Negative	Neutral	Positive
Baseline CNN	Good	Fair	Good
CNN + LSTM	Good	Fair	Good
RNN	Good	Good	Good
Advanced Model	Good	Fair	Good

Table 5. Confusion Matrix Analysis.

Conclusion and future enhancements

This paper presents an innovative method for emotion recognition utilising electroencephalogram (EEG) data, incorporating IndependentComponent Analysis (ICA) for artifact elimination and a hybrid metaheuristic optimisation algorithm (ABCGWO) for feature extract-ion. The principal findings indicate that IndependentComponent Analysis (ICA) proficiently eliminates noise from Electromyogram (EMG) and Electrooculogram (EOG) artifacts, thereby enhancing data quality for analysis. Furthermore, the hybrid ArtificialBee Colony-GreyWolf Optimisation (ABC-GWO) algorithm augments feature extraction, culminating in markedly improved emotion recognition accuracy, attaining up to 100% on the DEAP dataset and 99% on the SEED dataset. The methodology's resilience and generalisability are demonstrated by its performance on different datasets, indicating its potential for diverse applications in emotion recognition. The elevated accuracy rates indicate significant progress in the field, with methodology relevant to various areas necessitating precise signal processing and feature extraction, including human-computer interaction, mentalhealth monitoring, and adaptivelearning environments. The study recognises necessity for validation using larger and more diverse datasets, exploration of personalised models to accommodate individual EEG signal variations, optimisation for real-time applications, and integration with other modalities such as facial expressions and voice analysis to develop a comprehensive emotion recognition system. Thus, proposed methodology exhibits considerable improvements in emotion recognition accuracy, and by rectifying identified biases and limitations while investigating future research avenues, its relevance and influence can be further augmented with-in wider scope of emotion recognit-ion research.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on request.

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Declarations

Competing interests

The authors declare no competing interests.

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